Ryan Faatih Firdaus

Back-End & IoT Software Engineer
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SUMMARY

Back-End & IoT Software Engineer specializing in real-time IoT monitoring platforms and API integrations for large-scale on-premise deployments. Built a real-time monitoring platform serving 600+ devices and 500+ active users, the API integration layer for a separate 1,000-device on-premise project, and 3 C++ firmware projects for IoT devices. Experienced with Node.js, Laravel, Go, C++, Redis, MQTT, and Docker on Linux.

SKILLS

Languages: Go, C++, JavaScript, TypeScript, PHP, SQL
Back-End & Frameworks: Node.js, Laravel, React
Databases & Messaging: MySQL, PostgreSQL, Redis, Supabase, MQTT
Architecture & Integration: REST APIs, Webhooks, Microservices, Third-Party API Integration, IoT Telemetry, Embedded Firmware (C++)
Infrastructure & Tools: Docker, Linux, Nginx, Grafana, Git

EXPERIENCE

Software Engineer
PT Quanta Teknik Gemilang | Indonesia | Jan 2025 – Present

IoT Platforms, Firmware & Data Pipeline

- Built a real-time IoT monitoring platform ingesting live telemetry (5+ sensor metrics) from 2000+ devices over MQTT, serving 500+ active users.
- Redesigned the telemetry pipeline after hundreds of devices writing every MQTT reading to MySQL every 5 seconds saturated server CPU and memory: moved live data to Redis (latest value per device, refreshed every 15 seconds) and persisted to MySQL every 15 minutes, cutting database writes per device by 180x, CPU usage by ~98%, and memory usage by ~75%.
- Developed 3 IoT device firmware projects in C++ and improved several existing firmware, owning the full stack from device to back end to dashboard.
- Migrated a legacy PHP monitoring dashboard to a React front end and Laravel back end, improving maintainability.
- Building a Data Center Infrastructure Management (DCIM) platform in Go (in progress).

APIs & Integrations

- Built the API integration layer for a separate 1,000-device on-premise project, connecting field devices to the platform via REST APIs and webhooks.
- Designed and maintained public and internal REST APIs enabling interoperability between IoT devices, external platforms, and internal services.
- Developed 7 Node.js microservices for data processing pipelines, background jobs, and external system integrations, including a notification service that delivers alerts via webhooks and REST callbacks.

Infrastructure & Monitoring

- Containerized and deployed services with Docker on Linux servers, using Nginx as a reverse proxy to route traffic to each service.
- Set up Grafana dashboards and alerts to monitor server health and notify the team of issues such as storage running full.

Back-End Web Developer Intern
PT Origin Wiracipta Lestari | Indonesia | Jan 2024 – Mar 2024

- Built an inventory management system in PHP from scratch, with real-time stock tracking, full incoming/outgoing transaction history, and 50+ SKUs.
- Developed a production module covering the workflow from raw materials to finished goods, with automatic stock deduction at each stage.
- Designed the relational database schema to ensure data integrity and query performance across all modules.

EDUCATION

Bachelor of Information Systems
Universitas Islam Negeri Syarif Hidayatullah Jakarta | Sep 2021 – Dec 2025
GPA: 3.57 / 4.00